
Data Processing & Mining

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2025, Ukrainian Catholic University

No, as a Language Model I cannot depict you as a capybara



Dmytro Chaplynskyi

Developer, 20+ years of commercial experience; 15 years with Python.

10 years in the field of Ukrainian language processing (NLP); co-organizer of the [LANG-UK](#) public initiative.

Author/co-author of [10+ articles over](#) the last 3 years, including on the [UberText 2.0](#) and [NER UK 2.0](#) corpora.

Working on creating datasets, LLMs and machine translation.

Second-year graduate student at the Ukrainian Catholic University.

Fun fact about me



I'm on the right

Our topics

Can you guess the main word? Yeah, it's about...

→ **Data**

Why it's important, what we know about data in general?

→ **Data scraping**

How to create your own datasets?

→ **Data processing**

Why and how process your data?

Recipe of DL research:

Ideas

Data

Skills

Compute

Data is 10x more powerful than algorithms



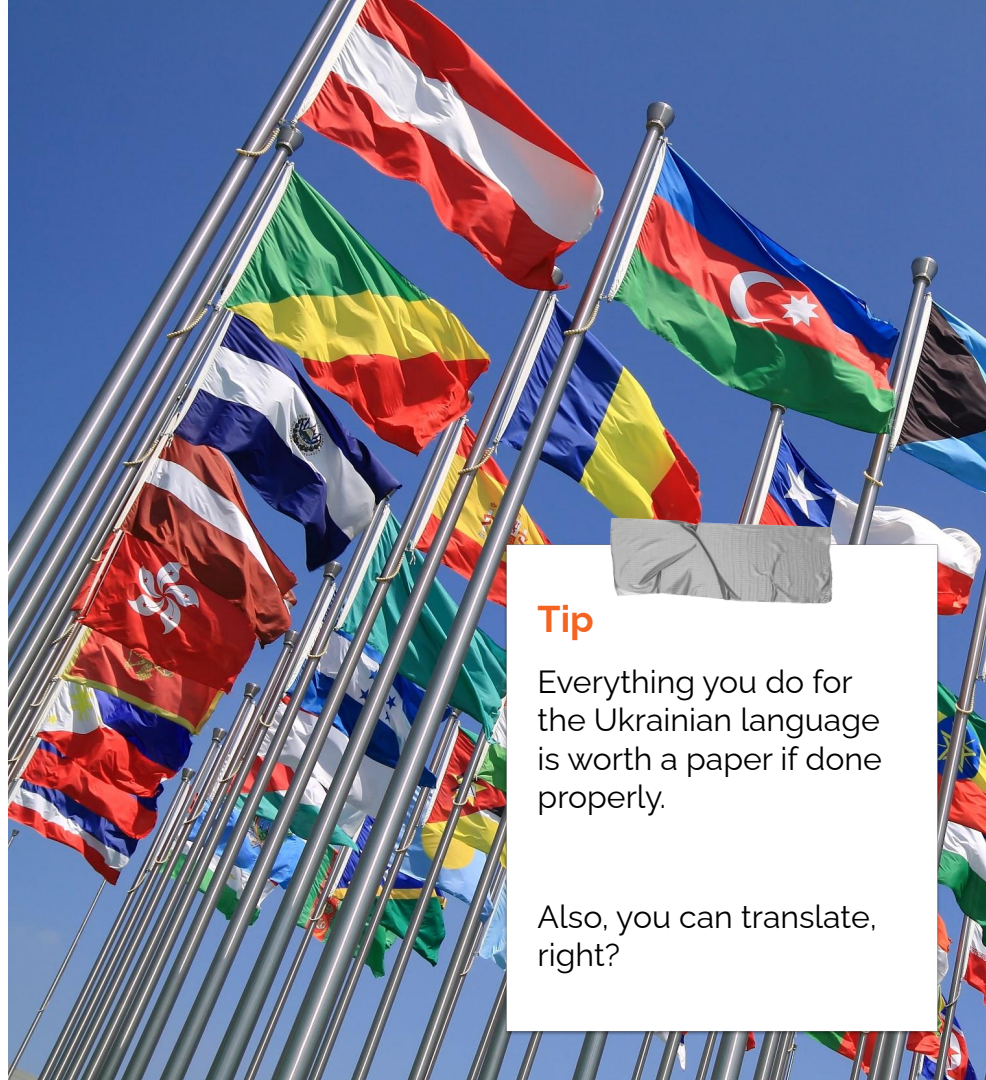
Tip

Watch his lecture **"The Unreasonable Effectiveness of Data"**

<http://youtu.be/yvDCzhbjYWs>

Not all the languages have
enough data for the **NLP**
tasks but we can fix it!

Просто візьміть старий
радянський...



Tip

Everything you do for
the Ukrainian language
is worth a paper if done
properly.

Also, you can translate,
right?



Dive into data

Data can be different, right?

→ **Open/proprietary**

Watch for the license, respect the ownership

→ **Structured/unstructured**

Know your tools, know how to introduce structure to the mess

→ **Poor quality/Good quality**

Oh gosh, that's hard, but we'll touch it too.

Licenses cheatsheet

License	Can I... (share / adapt / commercial)?	Must I... (attribute / share-alike / non-commercial)?	Good for	Official page
CCo 1.0	✓ / ✓ / ✓	No / No / No	Public-domain style release of data & metadata	Creative Commons
CC BY 4.0	✓ / ✓ / ✓	Yes / No / No	Open data with required attribution	Creative Commons
CC BY-SA 4.0	✓ / ✓ / ✓	Yes / Yes / No	Open data that must remain open under same terms	Creative Commons
CC BY-NC 4.0	✓ / ✓ / No (NC)	Yes / No / Yes	Community/academic sharing where commercial use is restricted	Creative Commons
ODC-By 1.0	✓ / ✓ / ✓	Yes / No / No	Databases needing attribution (EU/UK SGDR aware)	opendatacommons.org
ODbL 1.0	✓ / ✓ / ✓	Yes / Yes (DB SA) / No	Datasets where improved/redistributed DBs must stay open	opendatacommons.org
PDDL 1.0	✓ / ✓ / ✓	No / No / No	Putting a database fully in the public domain	opendatacommons.org
CDLA-Permissive 2.0	✓ / ✓ / ✓	Keep license text; no SA/NC	AI/ML-friendly permissive data sharing	CDLA
UK OGL 3.0	✓ / ✓ / ✓	Yes (attribution) / No / No	Government-published datasets (UK & interoperable)	The National Archives

Tip

Стаття 8. Об'єкти, що не охороняються авторським правом

[ЗАКОН УКРАЇНИ](#)
[Про авторське право і суміжні права](#)

Legal considerations

Respect licenses (for models, datasets, code, data sources)

Publish your work under permissive licenses when possible

Consult with lawyers

Accept feedback

A hand holding a smartphone, with a red background. The text is overlaid on the left side of the image.

Structure

Texts

Data sources

Gathering of data

Structure

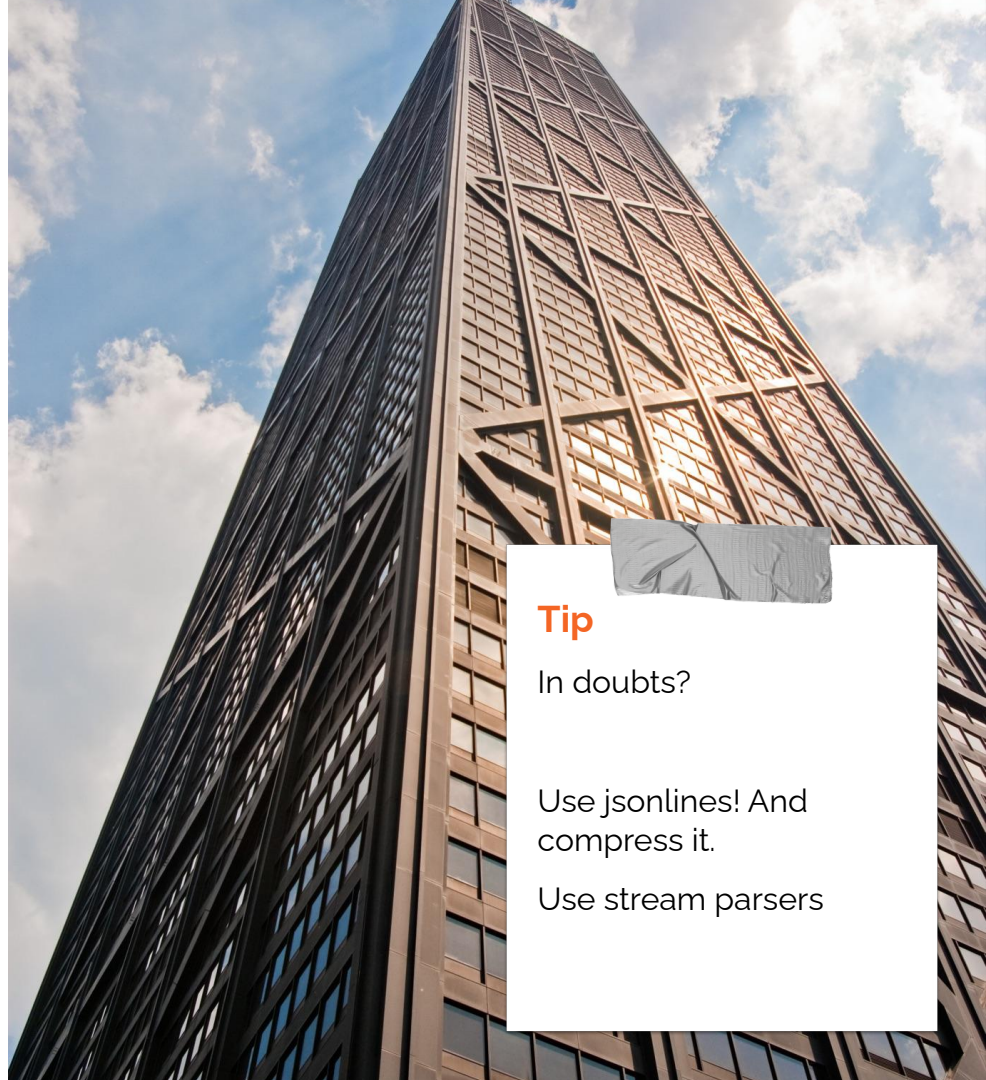
- Raw
- Semi-raw
- Tabular (csv, tsv, xlsx, dataframes)
- Nested (json, jsonlines, xml, yaml)

Tip

In doubts?

Use jsonlines! And compress it.

Use stream parsers





Raw and semi-raw

Internal structure of the texts.
Your ideas?

The need of models to extract implicit
structure

Little helpers of semi-structured texts
(can you name any?)

Tip

Learn from others

Preserve metadata as
much as possible

Eat healthy

Data model and annotation layers

Prepare it

Document it

Adhere to it

Sources for the data

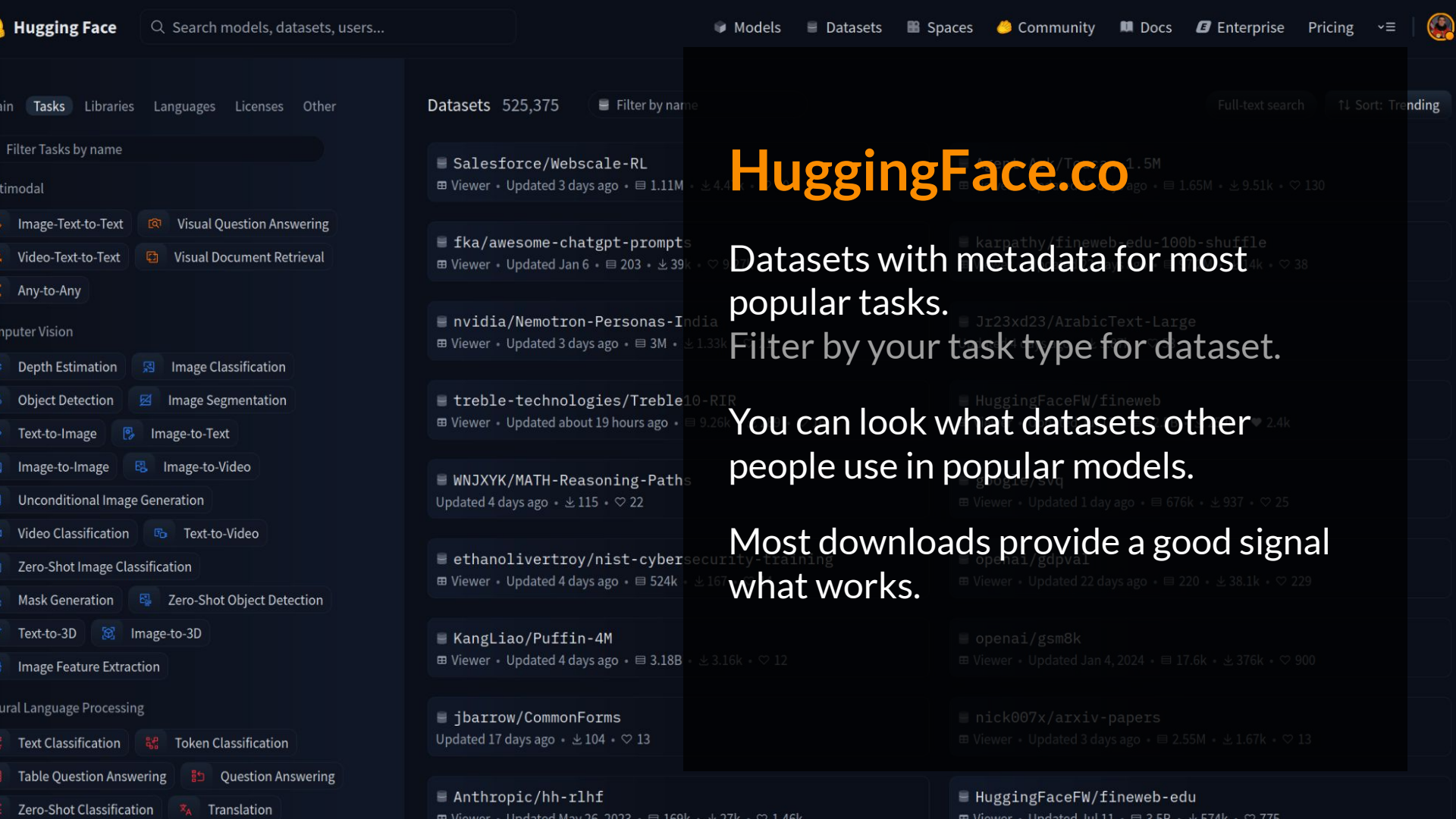
- APIs, downloadable datasets (REST, GraphQL)
- Wikidata (SparQL)
- Wikipedia (wtf_wikipedia, dumps)
- World Wide Web (scrapers, microformats)



Tip

Improve your
programming skills

Know your tools
(including CLI)

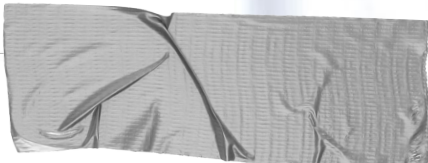


HuggingFace.co

Datasets with metadata for most popular tasks.
Filter by your task type for dataset.

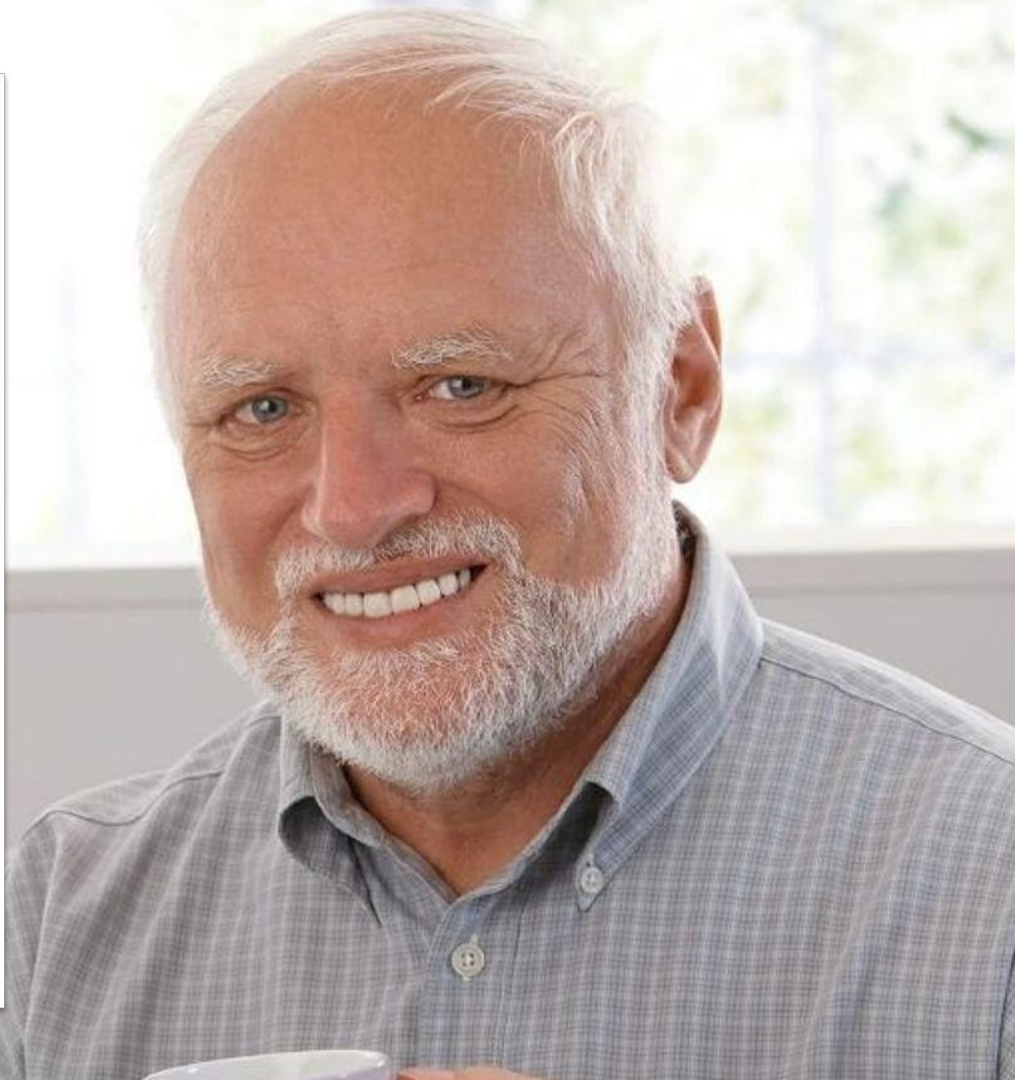
You can look what datasets other people use in popular models.

Most downloads provide a good signal what works.



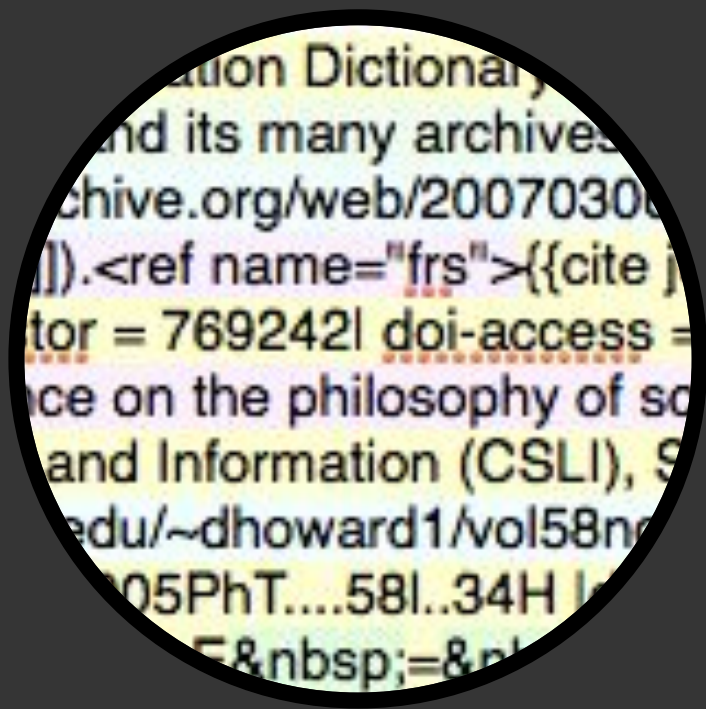
Scraping code of conduct

- Stay polite
- Don't crawl too harsh
- Respect copyrights and licenses
- Respect robots.txt when possible
- Cite your sources



—
Terribilis terra

Trying to make
sense out of
semi-structured
data!



Know your tools!

- Text editor is your main weapon
- Unix CLI tools, pipes, streams
- Regex
- JMESPath



Tip (repeated)

Improve your
programming skills

Know your tools
(including CLI)

Unix CLI tools

- Shortcuts for your terminal, shell and tools. History, navigation. PIPES
- screen, tmux, nohup, mc
- Monitoring: ps, top, htop, btop, du, ncd, nvidia-smi, iotop, pv
- Compression: bzip2/lbzip2, gzip, lzma, tar
- Search, filter and sort: grep, find, sort, uniq, sed
- wc, pv, watch, head, tail, split, jq, less, cat



No, seriously

- There is a tool for literally everything
- Find yourself a cheatsheet and print it
- Learn hotkeys
- Start using terminal if you don't
- Learn at least tmux
- Start using pipes

Regex

regex101.com

Beautiful visualisation

Different regex dialects

Explanation of everything

regexpal.com

When in hurry

pip install regex

import regex



jmespath

JMESPath is a query language for JSON.

```
people[?age > `20`].[name, age]
```

```
reservations[].instances[].[tags[?Key=='  
Name'].Values[] | [0], type, state.name]
```

```
locations[?state == 'WA'].name |  
sort(@)[-2:] | {WashingtonCities:  
join(', ', @)}
```

```
Q locations[?state == 'WA'].name | sort(@) | {WashingtonCities: join(', ', @)}
```

```
{  
  "locations": [  
    {"name": "Seattle", "state": "WA"},  
    {"name": "New York", "state": "NY"},  
    {"name": "Bellevue", "state": "WA"},  
    {"name": "Olympia", "state": "WA"}  
  ]  
}
```

Result

```
{  
  "WashingtonCities": "Bellevue, Olympia, Seattle"  
}
```

JMESPATH

Data annotation

- Understand what you tag, look for what the others did
- Find a tool ([brat](#), [LabelStudio](#), [MechanicalTurk](#), [Vulyk](#))
- Find a labor (do it yourself, ask friends, pay money, try twitter, fiverr, tinder)
- Prepare the data, the tagset, decide on the output formats
- Run, measure, repeat
- Edit, prepare, release
- Write a paper, seriously

NVIDIA NeMo Curator

GPU-accelerated data curation for training better AI models, faster. Scale from laptop to multi-node clusters with modular pipelines for text, images, video, and audio.

Part of the [NVIDIA NeMo](#) software suite for managing the AI agent lifecycle.

What You Can Do

Modality	Key Capabilities	Get Started
Text	Deduplication • Classification • Quality Filtering • Language Detection	Text Guide
Image	Aesthetic Filtering • NSFW Detection • Embedding Generation • Deduplication	Image Guide
Video	Scene Detection • Clip Extraction • Motion Filtering • Deduplication	Video Guide
Audio	ASR Transcription • Quality Assessment • WER Filtering	Audio Guide

Nemotron pipeline

- Language Identification + Unicode Fixing.
- Exact Deduplication.
- Fuzzy deduplication.
- Heuristic Filtering.
- Translate + Quality Classifier.
- Synthetic Data Generation for High-Quality data.

Heuristic Filtering Example

```
# the order they appear in this file. Each filter can be removed and re-order
#- name: nemo_curator.filters.heuristic_filter.NonAlphaNumericFilter
#  params:
#    max_non_alpha_numeric_to_text_ratio: 0.25
- name: nemo_curator.filters.heuristic_filter.SymbolsToWordsFilter
  params:
    max_symbol_to_word_ratio: 0.1
- name: nemo_curator.filters.heuristic_filter.NumbersFilter
  params:
    max_number_to_text_ratio: 0.15
- name: nemo_curator.filters.heuristic_filter.UrlsFilter
  params:
    max_url_to_text_ratio: 0.2
- name: nemo_curator.filters.heuristic_filter.WhiteSpaceFilter
  params:
    max_white_space_ratio: 0.25
```

Model-based processing

- You can use pretrained models or finetune custom ones to process your data
- LLM-based processing could be either too pricey or takes too long for locally deployed models
- You can estimate quality of the data using trained classifiers based on encoder models
 - e.g. Classification models, outputting a score from 0 to 1 for target aspect
 - For scaled inference you can use frameworks like CrossFit (NVIDIA)



Quality aspects

Transferred to Ukrainian from English:

- Mixtral-Edu - measure educational value of the text
- Nemotron-Edu - measure educational value of the text
- FastText - 0 - Common Crawl, 1 - Reddit ELI5.

Trained using Ukrainian datasets:

- Manipulative content detection.
- Alignment - Ukrainian

Context/Disinformation Alignment score.

propaganda - 0, factual claim - 1

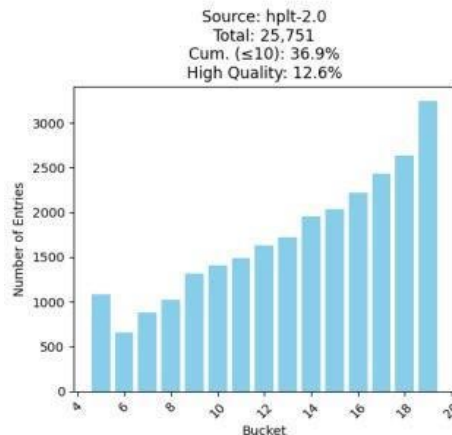
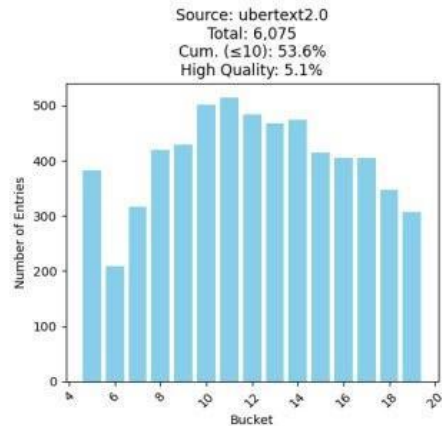
- Grammatical Error Correction.

You can then combine them together into one score using reference set..



Filtering example

- Bucketing means CDF binning.
- Then we calculate the max bucket for every text
- We perform manual sanity check for high and low scoring datapoints.
- We drop data that has a max score across all classifiers of 10 and lower to remove low-quality data.
- We split data into two buckets: bucket=19 is considered high-quality and data outside bucket 19 but with a high "formula" score
- everything else that's left is considered lower-quality
- Source dataset shrinks from 79B tokens to 30B tokens



My contacts

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<https://github.com/lang-uk/>

<https://huggingface.co/lang-uk>





Questions?